

End-to-End Process · Order-to-Cash (O2C)

PCF 3.5 Manage Sales Orders · PCF 4.4 Manage Logistics and Warehousing · PCF 9.4 Customer Credit, Invoicing & AR

View 1 — HITL Design

Pattern Type Legend

Decision 80% H at M3	Document 40% H	Transaction 35% H	Exception 82% H
--------------------------------	--------------------------	-----------------------------	---------------------------

O2C End-to-End Process — HITL Design · PCF-Derived Process Ownership

Sales PCF 3.5	Materials PCF 4.4	Finance PCF 9.4	Customer External participant
-------------------------	-----------------------------	---------------------------	---

HITL Design Insight

HITL Design Insight — O2C

4 Decision gates (80% Human at M3) are the irreducibly human layer — order acceptance, availability determination, and fulfillment routing carry commercial and customer authority that must remain human by design. **12 Transaction-pattern steps** (35% Human at M3) represent the AI absorption opportunity — fulfillment, invoicing, AR maintenance, payment processing, and cash application in rules-based sequence. **4 Exception flows** (82% Human) require human resolution — backorders, order inquiries, collections/dunning, and dispute resolution. O2C aggregate at M3: **52% Human / 48% AI** — grounded in APQC PCF SIPOC reference architecture and consistent with the enterprise-wide Human-AI convergence finding.

HITL Impact Analysis

H% = Human involvement at each maturity level · M3 = HITL Design Intent · ★ = Design target

Process Step (L3)	PCF	Pattern	M1 H%	M3 H% ★	Δ M1→M3	Opportunity
O2C End-to-End Process 3.5+4.4+9.4		Mixed	68%	52%	-20%	Significant
PCF 3.5 · Manage Sales Orders						
↳ Accept & Validate Sales Order	3.5.4	Decision	92%	80%	-12%	Retain Human
↳ Determine Availability	3.5.4	Transaction	60%	35%	-25%	High
↳ Determine Fulfillment Process	3.5.4	Decision	92%	80%	-12%	Retain Human
↳ Confirm Sales Order	3.5.4	Document	72%	40%	-32%	High
↳ Process Backorders & Updates	3.5.4	Exception	95%	82%	-13%	Retain Human
↳ Handle Order Inquiries	3.5.4	Exception	95%	82%	-13%	Retain Human
PCF 4.4 · Manage Logistics and Warehousing						
↳ Pick / Pack / Ship	4.4.3	Transaction	60%	35%	-25%	High
↳ Track Inventory Accuracy	4.4.3	Transaction	60%	35%	-25%	High
↳ Plan & Execute Outbound Delivery	4.4.4	Transaction	60%	35%	-25%	High
↳ Track Carrier Delivery Performance	4.4.4	Transaction	60%	35%	-25%	High
PCF 9.4 · Manage Customer Credit, Invoicing, and Accounts Receivable						
↳ Generate Invoice	9.4.2	Document	72%	40%	-32%	High

End-to-End Process · Order-to-Cash (O2C)

PCF 3.5 Manage Sales Orders · PCF 4.4 Manage Logistics and Warehousing · PCF 9.4 Customer Credit, Invoicing & AR

↳ Apply Taxes	9.4.2	Transaction	60%	35%	-25%	High
↳ Distribute Invoice	9.4.2	Transaction	60%	35%	-25%	High
↳ Maintain AR	9.4.3	Transaction	60%	35%	-25%	High
↳ Monitor Aging	9.4.3	Transaction	60%	35%	-25%	High
↳ Reconcile Accounts	9.4.3	Transaction	60%	35%	-25%	High
↳ Manage Collections / Dunning	9.4.4	Exception	95%	82%	-13%	Retain Human
↳ Resolve Disputes	9.4.4	Exception	95%	82%	-13%	Retain Human
↳ Process Customer Payment	9.4.5	Transaction	60%	35%	-25%	High
↳ Apply Cash	9.4.5	Transaction	60%	35%	-25%	High
External Participant · Customer						
↳ Submit Sales Order	Ext	Document	72%	40%	-32%	High
↳ Receive Order Confirmation	Ext	Transaction	60%	35%	-25%	High
↳ Receive Goods / Delivery	Ext	Transaction	60%	35%	-25%	High
↳ Receive Invoice	Ext	Transaction	60%	35%	-25%	High
↳ Submit Payment	Ext	Transaction	60%	35%	-25%	High
↳ Receive Payment Confirmation	Ext	Transaction	60%	35%	-25%	High

AI Capability Map — O2C

PCF 3.0 → 3.5 Market & Sell — Manage Sales Orders		PCF 4.0 → 4.4 Supply Chain — Manage Logistics and Warehousing		PCF 9.0 → 9.4 Financial Resources — Credit, Invoicing & AR	
3.5.4 Accept & validate sales orders	DECISION	4.4.3 Pick / pack / ship	TRANSACTION	9.4.2 Generate / distribute invoice	DOCUMENT
3.5.4 Determine availability	TRANSACTION	4.4.3 Track inventory accuracy	TRANSACTION	9.4.3 Maintain AR / monitor aging	TRANSACTION
3.5.4 Determine fulfillment process	DECISION	4.4.4 Plan & execute outbound delivery	TRANSACTION	9.4.3 Reconcile accounts	TRANSACTION
3.5.4 Confirm sales order	DOCUMENT	4.4.4 Track carrier delivery performance	TRANSACTION	9.4.4 Manage collections / dunning	EXCEPTION
3.5.4 Process backorders & updates	EXCEPTION			9.4.4 Resolve disputes	EXCEPTION
3.5.4 Handle order inquiries	EXCEPTION			9.4.5 Process payment / apply cash	TRANSACTION
M1: 76% M3: 57% M5: 30%		M1: 60% M3: 35% M5: 15%		M1: 68% M3: 48% M5: 20%	